

Analyzing Turkish Stock Market Behavior

Introduction and Motivation

The stock prices are simultaneously impacted by macroeconomic conditions, investor psychology, geopolitical uncertainty, monetary policy decisions, exchange-rate fluctuations, global market sentiment, and unforeseen external shocks, financial markets are among the most complex systems studied within data science. Financial markets behave stochastically, which means that uncertainty and unpredictability are essential elements of price fluctuations, in contrast to deterministic systems. Predicting stock returns is still one of the most challenging uses of statistical modeling and machine learning because of this extremely dynamic structure. The fact that relationships between variables continuously change over time is one of the most significant features of financial time-series data. Stock movements may be fairly predictable during times of economic stability, but during times of crisis, volatility can skyrocket and formerly stable relationships may completely collapse. The predictive power of traditional machine learning models is diminished as a result of nonlinear behavior, volatility clustering, heavy-tailed distributions, and quickly shifting market conditions.

Emerging markets are typically more sensitive to inflation, exchange-rate volatility, and macroeconomic shocks than established markets. The Turkish stock market offers a particularly intriguing setting for financial analysis. Investor behavior in emerging markets may also react more aggressively to political uncertainty and external economic developments, making Turkish stock returns highly volatile and difficult to model.

The primary motivation behind this project is to investigate whether statistical and machine learning methods can identify meaningful relationships between Turkish stocks and the broader market represented by the BIST100 index. Rather than focusing solely on stock prices themselves, this project attempts to understand the underlying financial behavior of Turkish stocks through exploratory data analysis, statistical hypothesis testing, regression analysis, and machine learning classification.

Understanding the difference between correlation and prediction is another important goal of this research. Although assets in financial markets frequently move in tandem under similar economic circumstances, a high correlation does not always indicate that future movements can be accurately predicted. Thus, the purpose of this study is to determine whether the relationships found during exploratory analysis indeed result in significant predictive performance.

The project integrates multiple concepts covered throughout the course lectures, including:

- data collection and preprocessing,
- exploratory data analysis,
- visualization interpretation,
- statistical inference,
- hypothesis testing,
- regression modeling,
- feature engineering,
- machine learning classification,
- model evaluation,
- and financial interpretation of predictive limitations.

The selected Turkish stocks represent different industries within Borsa Istanbul:

- THYAO (Turkish Airlines),
- GARAN (Garanti Bankası),
- ASELS (Aselsan),
- KCHOL (Koç Holding),
- TUPRS (Tüpraş),
- BIMAS (BİM).

To account for both domestic and international market impacts, the research included macro-level financial variables such as the BIST100 index, S&P500 index, and USD/TRY exchange rate in addition to individual stocks.

In contrast to the project's original iteration, the BIST100 index dataset—rather than the average return of particular stocks—was used to directly represent market performance. The average return of the chosen assets does not fairly reflect the

behavior of the Turkish market as a whole, so this change was made in response to teacher input. A more accurate and scholarly picture of market-wide circumstances can be obtained by using BIST100 as the benchmark.

The project attempts to answer four primary research questions:

- Do Turkish large-cap stocks systematically move together with the BIST100 market index?
- How do volatility, rolling beta values, and drawdowns evolve under changing market conditions?
- Are stock returns statistically different from overall market returns?
- Can machine learning algorithms predict the next-day direction of stock returns using historical financial indicators?

The project uses machine learning models like Logistic Regression, Random Forest, and Gradient Boosting in conjunction with financial time-series analysis to address these concerns. The ultimate goal is to comprehend the structure, constraints, and predictability of Turkish financial markets rather than only attaining high forecast accuracy.

Data Collection and Preprocessing

Using the Yahoo Finance API and the yfinance Python module, historical daily financial data was gathered. Both individual Turkish stock data and macro-level market indicators are included in the dataset, which spans the years 2018 to 2025. Numerous economically significant events, such as the post-pandemic recovery, inflationary times, exchange-rate volatility, monetary policy changes, and increased market volatility, are included in the chosen time period. As a result, the dataset offers a realistic setting for examining financial behavior in various market scenarios.

The chosen stocks were:

- THYAO.IS,
- GARAN.IS,
- ASELS.IS,
- KCHOL.IS,
- TUPRS.IS,
- BIMAS.IS,
- BIST100 Index (^XU100),
- S&P500 Index (^GSPC),
- USD/TRY Exchange Rate (USDTRY=X).

The following variables were collected for each asset:

- Open Price,
- High Price,
- Low Price,
- Close Price,
- Trading Volume.

Because adjusted closing prices take dividend adjustments and stock splits into account, they were utilized in return calculations. A more realistic depiction of actual investment performance over time can be obtained by using adjusted pricing.

Due to market vacations, halted trading, and asynchronous international trade schedules, missing observations are common in financial statistics. Preprocessing procedures were therefore required prior to statistical and machine learning analysis. Infinite or invalid observations were cleaned, trading dates were aligned across all datasets, and missing values were eliminated or forward-filled as necessary. Synchronizing trading dates was particularly important because Turkish markets and international markets such as the S&P500 do not always operate on identical trading calendars. Without date alignment, statistical calculations and regression results could become misleading.

Daily percentage returns were calculated using:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

where:

- R_t represents daily return at time t ,
- P_t represents current closing price,
- P_{t-1} represents previous closing price.

The daily returns offer a normalized depiction of stock performance and enable meaningful comparison across assets with varying price levels, they were chosen over raw prices. Using percentage returns instead of raw prices also reduces scale dependency between assets and allows statistical relationships to be interpreted more consistently.

Additionally, feature engineering was used to enhance predictive power. Market returns, momentum-related variables, rolling average returns, rolling volatility measures, and lagged returns were produced. Rolling volatility measures were meant to depict dynamic market risk and volatility clustering, whereas lagged return features were meant to reflect short-term temporal dependencies in financial behavior. Lagged returns are especially important in time-series analysis because they attempt to capture whether previous market movements contain information about future price behavior.

Exploratory Data Analysis

To comprehend the statistical structure of Turkish stock returns and their connection to market-wide factors, exploratory data analysis, or EDA, was carried out. Instead of just showing raw data, each figure should address a particular analytical question in accordance with the visualization concepts covered in the course lectures. As a result, each visualization employed in this project was chosen to examine a specific financial feature, such as market dependency, volatility persistence, downside risk, or asset interdependence.

The chosen Turkish stocks differed significantly in terms of return, volatility, and downside risk, according to descriptive statistics research. Among the companies examined, ASELS generated the greatest annualized return, demonstrating robust long-term growth performance during the chosen time frame.

However, ASELS also exhibited elevated volatility levels, suggesting that higher returns were associated with higher investment risk. THYAO demonstrated both strong returns and substantial downside exposure. The stock experienced the largest maximum drawdown among all analyzed assets, indicating that airline-sector companies are highly sensitive to macroeconomic instability and global uncertainty. BIMAS displayed comparatively lower volatility and smaller drawdown values, suggesting that consumer retail companies behave more defensively during periods of economic stress.

Stock returns were not regularly distributed, according to skewness and kurtosis statistics. Asymmetric behavior and large tails, which are typical features of financial time-series data, were seen in a number of return distributions. This result is significant since Gaussian return distributions are assumed by many traditional statistical and financial models. Predictive modeling and statistical inference become more difficult when this assumption is broken because extreme observations occur more frequently than expected under normal distribution assumptions.

To look at the connections between Turkish stocks and the BIST100 market index, a correlation heatmap was created. The following methods were used to calculate Pearson correlation coefficients:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \quad r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

where:

- x_i represents stock returns,
- y_i represents market returns,
- $\bar{x}$ represents mean stock return,
- $\bar{y}$ represents mean market return.

Strong positive correlations between the majority of the chosen firms and the BIST100 index were found in the analysis, suggesting that Turkish large-cap stocks typically move in tandem during similar market conditions. One of the greatest associations between KCHOL and BIST100 indicated a significant level of systematic market risk exposure. This suggests that macroeconomic factors and general market sentiment have a greater influence on KCHOL price behavior than firm-specific factors. Additionally, THYAO and GARAN showed comparatively high market dependence, indicating how sensitive the banking and aviation industries are to changes in the economy. BIMAS yielded relatively lower correlation values, indicating that businesses in the retail sector might be more resilient in times of market volatility.

This stage's key finding is that predictability is not always implied by substantial correlation. Although stocks move together with market conditions, these relationships may still fail to produce reliable forecasts for future returns because financial markets contain substantial randomness and rapidly changing external influences.

Rolling volatility analysis was conducted to examine whether market risk remains stable over time or exhibits clustering behavior. Rolling volatility was calculated using:

$$\sigma_t = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (R_i - \bar{R})^2}$$

where:

- σ_t represents rolling volatility,
- R_i represents daily returns,
- $\bar{R}$ represents average return,
- n represents rolling window size.

Periods of volatility clustering were evident in the rolling volatility graphs, where high-volatility periods tended to be followed by more high-volatility periods. This tendency implies that financial uncertainty does not disappear immediately. Rather, market shocks cause ongoing risk conditions as they spread across time. During periods of economic instability, THYAO and GARAN performed notably high

volatility spikes, but BIMAS showed comparatively smoother volatility behavior. This result is entirely in line with the financial time-series literature, which views volatility clustering as one of the key traits of stock return behavior. The persistence of volatility also suggests that market shocks influence investor behavior for extended periods rather than affecting prices only temporarily.

Additionally, rolling beta analysis was used to track changes in portfolio sensitivity to the BIST100 index over time. The following methods were used to compute beta values:

$$\beta = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)} \quad \beta = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

where:

- R_i represents portfolio returns,
- R_m represents BIST100 returns,
- $\text{Cov}(R_i, R_m)$ represents covariance between portfolio and market returns,
- $\text{Var}(R_m)$ represents market return variance.

Rolling beta values showed significant fluctuations over time, suggesting that market dependency is not continuous. Beta values rose dramatically during times of financial instability, indicating that portfolio behavior became more reliant on mood in the market as a whole rather than firm-specific data. This result shows how systematic market risk predominates in Turkish equity markets. During highly uncertain periods, investors tend to react more strongly to market-wide developments than to company-specific information, causing correlations and beta values to rise simultaneously.

To assess exposure to downside risk, maximum drawdown analysis was carried out. Among all the companies examined, THYAO had the most drawdown, indicating significant short-term losses during challenging market conditions. BIMAS generated relatively lower drawdowns, supporting the notion that businesses in the retail sector might act more defensively during recessions. This analysis shows that high-return assets frequently carry significant short-term losses and emphasizes the significance of downside risk management. Large drawdowns

are especially important for investors because temporary losses may strongly affect investor psychology and long-term portfolio stability.

Hypothesis Testing and Statistical Inference

To determine whether Turkish stock returns performed significantly differently from BIST100 market returns, statistical hypothesis testing was carried out. Hypothesis testing necessitates defining null and alternative hypotheses, choosing significance levels, computing test statistics, calculating p-values, rejecting or failing to reject the null hypothesis, and finally interpreting results within a financial context, according to the statistical inference framework covered in the lectures.

The first hypothesis looked at whether there was a substantial difference between the mean daily return of Turkish equities and the mean daily return of the BIST100 index.

The null and alternative hypotheses were defined as:

$H_0: \mu_{\text{stock}} = \mu_{\text{BIST100}}$

$H_A: \mu_{\text{stock}} \neq \mu_{\text{BIST100}}$

Since stock returns and BIST100 returns were observed on the same trading dates, the observations were dependent rather than independent. Therefore, paired t-tests were conducted instead of independent two-sample tests. The paired t-test statistic was calculated using:

$$t = \frac{\bar{d} - \mu_d}{s_d / \sqrt{n}}$$

where:

- $\bar{d}$ represents the mean difference between paired observations,
- s_d represents the standard deviation of differences,
- n represents the number of paired observations.

Most p-values exceeded the 0.05 significance level, meaning that the null hypothesis could not be rejected for most analyzed stocks. Although several stocks numerically outperformed the market during certain periods, these differences were not statistically stable over time. This suggests that large-cap Turkish stocks remain heavily influenced by systematic market conditions represented by the BIST100 index. The inability to reject the null hypothesis also supports the idea that outperforming the market consistently is statistically difficult, particularly within highly interconnected financial systems.

Jarque-Bera tests were applied to examine whether stock returns followed normal distributions. The Jarque-Bera statistic evaluates deviations from normality by jointly measuring skewness and kurtosis behavior. Extremely small p-values were obtained for all analyzed assets, leading to rejection of the null hypothesis of normality. This finding indicates that Turkish stock returns exhibit heavy tails, skewness, abnormal fluctuations, and extreme observations. The rejection of normality is particularly important because many classical financial models assume normally distributed returns. Heavy-tailed distributions also help explain why machine learning models struggle to accurately predict extreme market movements because abnormal observations may distort learning behavior.

Augmented Dickey-Fuller (ADF) tests were conducted to evaluate stationarity behavior. The ADF test examines whether a time series contains a unit root, which would indicate non-stationarity. All analyzed return series produced highly negative ADF statistics and extremely small p-values, leading to rejection of the null hypothesis of non-stationarity. This indicates that daily stock return series are stationary and therefore suitable for time-series machine learning analysis. Stationarity is essential because non-stationary data may generate unstable statistical relationships and misleading predictive performance. Stable statistical properties over time improve the reliability of regression and classification models.

Regression Analysis

Regression analysis was performed to quantify the relationship between portfolio returns and macroeconomic indicators. The linear regression model is represented as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n + \varepsilon$$

where:

- Y represents portfolio return,
- $X_1, X_2, \dots, X_n$ represents explanatory variables,
- $\beta_1, \beta_2, \dots, \beta_n$ represents regression coefficients,
- ε represents the error term.

The BIST100, S&P500, and USD/TRY returns were among the independent variables. BIST100 returns had the most explanatory power over portfolio behavior, according to the regression results. This result demonstrates that systemic market risk and domestic market conditions have a significant impact on Turkish stocks. Additionally, S&P500 returns showed a moderate explanatory power, indicating that Turkish stocks are somewhat influenced by the mood in the global market. Particularly during times of increased macroeconomic volatility, exchange-rate fluctuations showed reduced but still significant correlations with portfolio results.

The predominant performance of BIST100 coefficients implies that, in the examined portfolio, systematic market exposure has a greater impact than firm-specific impacts. This study supports previous rolling beta and correlation findings that showed significant market dependency among Turkish large-cap stocks. The regression findings also demonstrate that macroeconomic conditions dominate many short-term portfolio movements within Turkish equity markets.